

Channel B: the noisy saddle-node-on-circle and the $2/3$ phase exponent

bifurcation-specificity of the folded-cycle phase channel: SNIC vs. fold

Solomon Williams
University of Edinburgh

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Abstract

We finish the Channel-B (phase) inner analysis and pin its bifurcation-specific exponent. When the limit cycle is destroyed by a *fold of cycles* (finite period), the phase velocity is bounded below, the iPRC is bounded, and the phase diffusion is regular ($D_\phi \sim \sigma^2$): Channel A (amplitude escape, $\sigma_*^A \sim \varepsilon_2^{1/2}$) governs. When it is destroyed by a *SNIC* (saddle-node on the invariant circle, diverging period), the saddle-node lives *in the phase*: the inner phase equation is the noisy saddle-node-on-circle, which rescales to the parameter-free canonical form $du = \frac{1}{2}u^2 d\tau + dW_\tau$ at the inner scale $\theta \sim \sigma^{2/3}$. Hence the noise-induced rotation rate and phase diffusion scale as $\sigma^{2/3}$ —verified, $\omega \sim \sigma^{0.668}$ —with the iPRC diverging as $(\mu - \mu_c)^{-1/2}$. The clean exponent is $\boxed{2/3}$; a previously reported 0.83 is the slowly-converging pre-asymptotic effective *diffusion* exponent, which shares the same $2/3$ root. This is a scaling result (matched asymptotics + the canonical reduction + numerics), honestly weaker than the Tracy–Widom rigour of Channel A.

1 The two channels and the inner phase equation

Near the fold the oscillator carries an amplitude r and a phase θ . *Channel A* is amplitude escape across the canard (the Tracy–Widom story, rigorous). *Channel B* is the accumulation of phase noise, with effective amplitude weighted by the infinitesimal phase-response curve (iPRC) $Z(\theta) = \omega/v(\theta)$, where $v(\theta) = d\theta/dt$ is the phase velocity and ω the mean frequency. The bifurcation that destroys the cycle decides which channel is singular.

Fold of cycles (finite period). Two limit cycles collide and annihilate at *finite* period: the saddle-node is in the *amplitude* direction ($dr = (r^2 - y) dT + \eta dB$, Channel A). The phase velocity stays bounded below, $v(\theta) \geq v_{\min} > 0$, so the iPRC is bounded and the phase obeys a regular diffusion $d\theta = v dt + \sigma Z dW$ with $D_\phi \sim \sigma^2$. Channel B is regular; $\sigma_*^A \sim \varepsilon_2^{1/2}$ (exponent $\frac{1}{2}$) governs, and “A preempts B”.

SNIC (diverging period). A saddle-node forms *on* the invariant circle: a fixed point appears in the phase, the period diverges, and the saddle-node is in the *phase* direction. The Adler normal form is $d\theta/dt = \mu - \cos\theta$, with the SNIC at $\mu_c = 1$; near the bottleneck $\theta \approx 0$,

$$\frac{d\theta}{dt} = (\mu - 1) + \frac{1}{2}\theta^2 + \sigma \xi. \quad (1)$$

Now the phase velocity *vanishes* at criticality, and Channel B is singular.

2 The 2/3 exponent

Proposition 1 (noisy saddle-node-on-circle reduction). *At criticality $\mu = \mu_c$, the bottleneck (1) rescales by $\theta = \sigma^{2/3}u$, $t = \sigma^{-2/3}\tau$ to the parameter-free canonical noisy saddle-node*

$$du = \frac{1}{2}u^2 d\tau + dW_\tau.$$

Consequently the bottleneck passage time is $\sim \sigma^{-2/3}$, and—since the bottleneck dominates the period—the noise-induced rotation rate and phase diffusion scale as

$$\omega \sim \sigma^{2/3}, \quad D_\phi \sim \sigma^{2/3}.$$

Proof. Substituting $\theta = \sigma^{2/3}u$, $t = \sigma^{-2/3}\tau$ into (1) at $\mu = \mu_c$: $d\theta = \sigma^{2/3}du$; $\frac{1}{2}\theta^2 dt = \frac{1}{2}\sigma^{4/3}u^2 \cdot \sigma^{-2/3}d\tau = \sigma^{2/3}\frac{1}{2}u^2 d\tau$; and the noise $\sigma dW = \sigma \cdot \sigma^{-1/3}dW_\tau = \sigma^{2/3}dW_\tau$ (since $dW \sim (dt)^{1/2} = \sigma^{-1/3}(d\tau)^{1/2}$). Dividing through by $\sigma^{2/3}$ gives $du = \frac{1}{2}u^2 d\tau + dW_\tau$, free of σ . Hence the inner time is $\sigma^{-2/3}$ (one $O(1)$ unit of τ), the bottleneck escape time is $\langle T \rangle \sim \sigma^{-2/3}$, and $\omega = 2\pi/\langle T_{\text{period}} \rangle \sim \sigma^{2/3}$ since the bottleneck dominates the period. The diffusion of the resulting renewal rotation inherits the same scale, $D_\phi \sim \sigma^{2/3}$. $\square$

Verification. Simulating $d\theta = (1 - \cos \theta)dt + \sigma dW$ and measuring $\omega = \langle \theta(T) \rangle / T$, $D_\phi = \text{Var}(\theta(T)) / 2T$:

$$\omega \sim \sigma^{0.668} \text{ (}\sigma^{0.671} \text{ on the small-}\sigma \text{ half),} \quad D_\phi \sim \sigma^{0.73-0.78},$$

with $D_\phi/\omega \approx 1.1$ roughly constant. The drift pins the exponent at 2/3 to three digits; the diffusion's effective exponent drifts from ≈ 0.78 toward 2/3 as $\sigma \rightarrow 0$ (a known pre-asymptotic feature of giant diffusion near criticality [1]). *A previously reported 0.83 is this pre-asymptotic diffusion estimate, not a distinct exponent.*

3 Bifurcation-specificity

Lemma 1 (iPRC dichotomy). *For the Adler oscillator the iPRC peak is $Z_{\max} = \omega/v_{\min} = \sqrt{\mu^2 - 1}/(\mu - 1) \sim (\mu - \mu_c)^{-1/2}$, which diverges at the SNIC; for a fold of cycles $v_{\min} > 0$ stays bounded, so $Z_{\max} = O(1)$.*

The exponent is therefore set by *where the saddle-node lives*:

	saddle-node in	phase channel
fold of cycles	amplitude (r); period finite	regular $D_\phi \sim \sigma^2$; Channel A $\frac{1}{2}$ governs
SNIC	phase (θ); period $\rightarrow \infty$	singular ω , $D_\phi \sim \sigma^{2/3}$; iPRC $\sim (\mu - \mu_c)^{-1/2}$

The folded *limit cycle* of JKK is a fold of cycles, so its noise response is governed by Channel A ($\sigma_*^A = 2\sqrt{\pi}\sqrt{\varepsilon_2}\sqrt{ac/b}$, Tracy–Widom inner exit). A SNIC realisation instead exposes the phase channel, with the 2/3 signature. This is the precise sense of Channel B's bifurcation-specificity.

4 Honest scope

What is established. The inner phase equation in both cases; the canonical reduction (Proposition 1) is exact and gives the clean exponent 2/3, *verified* in the drift to three digits; the iPRC dichotomy (Lemma 1) is closed-form; and the bifurcation-specificity is pinned to the location of the saddle-node.

What remains soft. Unlike Channel A’s exact Tracy–Widom law, Channel B is a *scaling* result: the canonical noisy saddle-node $du = \frac{1}{2}u^2 d\tau + dW$ is exact, but its first-passage statistics (the prefactor, the precise diffusion constant, the slowly-converging D_ϕ exponent) are not in closed form, and the full *swept* SNIC passage (with the slow drift through μ_c compounding the bottleneck) is at the scaling level only. The honest statement is: Channel B’s mechanism and clean exponent are identified; its rigour is one rung below Channel A’s.

References

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- [2] G.B. Ermentrout, D. Terman, *Mathematical Foundations of Neuroscience*, Springer (2010) — iPRC, SNIC, type-I excitability.
- [3] S. Jelbart, C. Kuehn, N. Kuntz, arXiv:2208.01361 (2024).
- [4] Companion: `PathA_endgame_writeup` (Channel A / Tracy–Widom), `ShootingCharacterisation_proof`.